QML Lab • research project 08

Hybrid Modulation Classifier

Recognising BPSK, QPSK and 16-QAM signals with a hybrid quantum–classical network.

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Quadrant	CQ (classical data, quantum processor)	**Track**	Hybrid
Difficulty	Medium	**Tools**	PennyLane, PyTorch, NumPy
Builds on	Module 7 of the QML Lab	**Workspace**	`qml_projects/hybrid-modulation-classifier/`

Summary. Automatic modulation classification identifies how a received radio signal was modulated, a basic task in cognitive radio and spectrum monitoring. This project builds a hybrid network that feeds statistical signal features into a small quantum layer, and compares it with classical models across signal-to-noise ratios. It connects the QML Lab directly with communications engineering.

1 Background and key concepts

1.1 Signal model

After down-conversion, a received frame is a sequence of complex baseband samples

$$r[k] = s[k]\, e^{j(2\pi \Delta f\, k + \phi)} + w[k], \qquad \mathrm{SNR} = \frac{\mathbb{E}|s[k]|^2}{\mathbb{E}|w[k]|^2}, \tag{1}$$

where $s[k]$ are symbols from a constellation (BPSK, QPSK, 16-QAM), Δf a small frequency offset, ϕ an unknown phase and $w[k]$ complex Gaussian noise.

1.2 Cumulant features

Higher-order cumulants are classic features for modulation recognition (Swami and Sadler, 2000). With $C_{20} = \mathbb{E}[r^2]$ and $C_{21} = \mathbb{E}|r|^2$,

$$C_{40} = \mathbb{E}[r^4] - 3C_{20}^2, \qquad C_{42} = \mathbb{E}|r|^4 - |C_{20}|^2 - 2C_{21}^2. \tag{2}$$

Normalised by C_{21}^2, they take characteristic noise-free values, for example $\tilde{C}_{42} = -2$ (BPSK), -1 (QPSK) and -0.68 (16-QAM), and Gaussian noise drives all cumulants above second order towards zero. They are compact (a handful of numbers per frame), which suits a few-qubit layer.

1.3 Learned features

Deep networks can also learn directly from raw I/Q samples: convolutional networks (O'Shea et al., 2016) and residual networks on over-the-air data (O'Shea et al., 2018) are strong baselines, especially at low SNR.

1.4 Hybrid model

As in Module 7, the hybrid network is features → Linear → 4-qubit layer → Linear → softmax, trained end to end in PyTorch (Paszke et al., 2019); the quantum layer returns one $\langle Z \rangle$ per qubit.

2 Research questions

The project is open-ended: these questions are the starting point, not the finish line. Add new ones to the README and the journal as they come up.

RQ1 At which SNR does each model break down, and does the hybrid model degrade differently from classical ones?

RQ2 Which cumulant features suit a 4-qubit layer best, and does feature choice matter more than the model?

RQ3 Does the quantum layer make the model more or less robust to phase and frequency offsets?

3 Method

3.1 Data

Synthetic frames generated with NumPy: 128 samples per frame, random symbols, random phase, small frequency offsets, SNR from -5 to 20 dB in 5 dB steps, 1000 frames per class and SNR. Features: normalised $|C_{40}|$, $|C_{42}|$, C_{20} and amplitude statistics (6–8 numbers), plus raw I/Q for the CNN baseline.

3.2 Models

Hybrid network with $n = 4$ qubits and 2 layers of basic entanglers between classical linear layers; cross-entropy loss, Adam, 30 epochs. A version trained per SNR and one trained on all SNRs together.

3.3 Baselines

MLP on the same cumulant features with matched parameter count; a decision tree with cumulant thresholds in the spirit of Swami and Sadler (2000); a small CNN on raw I/Q samples (O'Shea et al., 2016).

3.4 Metrics

- Accuracy versus SNR curve for every model.
- Confusion matrices at low, medium and high SNR.
- Parameter counts and training time.

4 Planned experiments

Each experiment becomes one `[[experiments]]` entry in the workspace, so its runs, parameters and metrics are stored and compared in the portal.

ID	Experiment	What varies	Main metrics
E1	Three classes	cumulant features, all SNRs	accuracy
E2	SNR curve	SNR $-5..20$ dB	accuracy per SNR
E3	Qubit count	$n \in \{2, 4, 6\}$	accuracy, parameters
E4	Impairments	phase and frequency offset	robustness
E5	More classes	add 8-PSK, 64-QAM	confusion matrix

Table 1: Experiment plan. Results go into the journal and the results log (section 9).

5 Running it in the QML Lab

The workspace lives in `qml_projects/hybrid-modulation-classifier/`. Scripts import three helpers from `qml_lab/plots.py`:

- `params()` reads the parameters from the portal's run form (defaults when run by hand);
- `save()` stores figures with the run;
- `log_metrics()` records numbers that appear in the run table.

```
qml_projects/hybrid-modulation-classifier/
  project.toml          metadata + experiments
  README.md             goal, open questions, ideas
  experiments/          runnable scripts
  docs/main.tex         this document
```

First experiment: This experiment is *planned*: write the script, add the block below to `project.toml`, then run `python manage.py sync_projects`.

```
[[experiments]]
slug = "three-class"
title = "BPSK / QPSK / 16-QAM from cumulant features"
script = "experiments/e01_three_class.py"
figure = "accuracy_vs_snr.png"

[experiments.params]
n_qubits = 4
n_layers = 2
frame_len = 128
epochs = 30
snr_min = -5
snr_max = 20
seed = 0
```

```
# from the portal folder: pick up changes to project.toml
python manage.py sync_projects
cd qml_projects/hybrid-modulation-classifier
# run by hand with the default parameters
```

```
PYTHONPATH=../../qml_lab python experiments/e01_three_class.py
```

6 Quality bar

- A classical baseline with a similar number of trainable parameters, tuned with the same care (Bowles et al., 2024).
- At least five random seeds per configuration, reported as mean $\pm$ standard deviation.
- How results change with the number of qubits.
- At least one run under a realistic noise model.
- Always show accuracy as a function of SNR; a single averaged number hides where models fail.

7 Open directions

- Quantum convolution directly on raw I/Q windows.
- Quantum kernel SVM on cumulant features (connects to Projects 01 and 02).
- Real recordings from public radio data sets.

8 Risks and limitations

- At high SNR cumulants make the task nearly linear, so every model reaches 100 % and differences vanish.
- At very low SNR all models fail; the interesting region is narrow.
- The CNN sees more information (raw samples); compare like with like as well.

9 Results log

Summarise findings here as they become stable; day-to-day notes belong in the portal journal.

Date	Run	Finding	Next step

References

Bowles, J., Ahmed, S. and Schuld, M. (2024) 'Better than classical? The subtle art of benchmarking quantum machine learning models', Preprint, arXiv:2403.07059. Available at: https://arxiv.org/abs/2403.07059.

O'Shea, T. J., Corgan, J. and Clancy, T. C. (2016) 'Convolutional radio modulation recognition networks', in *Engineering Applications of Neural Networks (EANN 2016)*, Communications in Computer and Information Science, vol. 629, Cham: Springer, pp. 213–226. Available at: https://doi.org/10.1007/978-3-319-44188-7_16.

O'Shea, T. J., Roy, T. and Clancy, T. C. (2018) 'Over-the-air deep learning based radio signal classification', *IEEE Journal of Selected Topics in Signal Processing*, 12(1), pp. 168–179. Available at: https://doi.org/10.1109/JSTSP.2018.2797022.

Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J. et al. (2019) 'PyTorch: An imperative style, high-performance deep learning library', in *Advances in Neural Information Processing Systems 32 (NeurIPS 2019)*, pp. 8024–8035.

Swami, A. and Sadler, B. M. (2000) 'Hierarchical digital modulation classification using cumulants', *IEEE Transactions on Communications*, 48(3), pp. 416–429. Available at: https://doi.org/10.1109/26.837045.